

Erdős Problem #944

EP-944

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

A critical vertex, edge, or set of edges, is one whose deletion lowers the chromatic number.

Let $k \geq 4$ and $r \geq 1$. Must there exist a graph G with chromatic number k such that every vertex is critical, yet every critical set of edges has size $> r$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>